

NICHOLAS W. LUNDGREN

Physicist & Computational Materials Scientist | Physical Chemistry PhD Candidate

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PROFESSIONAL SUMMARY

PhD candidate in Physical Chemistry (UC Davis, expected Dec 2026) specializing in thermal transport physics, computational materials science, and Machine-Learning (ML)-accelerated atomistic simulation. Designed novel modeling approaches for amorphous and 2D semiconductor systems. Co-developer of kALDo, an open-source thermal transport library. Experienced mentoring students and translating advanced physics to cross-disciplinary audiences.

CORE COMPETENCIES

- Heat transport & phonon physics
- Lattice dynamics & molecular dynamics
- ML interatomic potentials (NEP)
- DFT+DFPT (Quantum Espresso)
- Atomistic simulation (LAMMPS, GPUMD, DLPOLY)
- Python 3 (expert), Fortran
- HPC systems & workflow automation (SLURM)
- Open-source software development (kALDo)
- Data analysis, visualization & scientific computing
- Technical writing & cross-discipline communication
- Microsoft Office (Word, Excel, Powerpoint)

EXPERIENCE

Graduate Researcher — Computational Materials Science | *University of California, Davis* Sep 2020 – Present

- Studied thermal transport in low-symmetry and amorphous semiconductors using molecular and lattice dynamics with ML potentials. Resulting in a PRB publication and another in submission (to PRB)
- Statistical analysis of experimental and simulation datasets for comparison, characterizing distributions, variance, and time correlations (heat-flux) using custom-written tools and data structures.
- Trained and validated ML interatomic potential models (NEP framework) for 2D silicene on GPU hardware; used clustering analysis for dataset curation and bench marked force-field accuracy and transferability against DFT.
- Co-developed kALDo, an open-source Python thermal transport library: implemented full new features (NAC, Parallelization), maintained CI/CD pipelines, and reviewed code/model contributions across a multi-author repository. Published in JAP (2020) and CPC (2026).
- Mentored an undergraduate researcher through an independent computational project crushing nanotubes for thermal transport simulations (ongoing).

Teaching Assistant | *University of California, Davis, Dept. of Chemistry* 2020 – 2026

- Educated undergraduate and graduate students on advanced topics spanning quantum chemistry, NMR spectroscopy, thermodynamics, organic chemistry, and kinetics. Analogous to educating non-technical communities on scientific concepts and statistics.
 - Developed instructional materials (slides) and held office hours supporting 700+ students per quarter.
 - Recognized with a Best Teaching Assistant nomination twice.
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EDUCATION

PhD in Chemistry (Physical Chemistry) | *University of California, Davis* Dec 2026 (Expected)

Research: Thermal transport in semiconductors & disordered materials; ML-accelerated atomistic simulation; open-source scientific software. GPA: 3.75

BS in Chemical Physics, Highest Honors (Summa Cum Laude) | *University of California, Davis* June 2020

Thesis: The Effects of Germanium Concentration on the Thermal Transport of Amorphous Silicon. GPA: 3.61

PUBLICATIONS & PRESENTATIONS

- Thermal Transport in Amorphous Carbon from Crushed Fullerenes — Drafted
- Layer-Stacking and Strain Effects on Heat Transport in Silicene — Submitted (July 2026)
- kALDo 2.0: Thermal Transport from ML Potentials — **Computer Physics Comms.** (2026)
- Layer-Stacking and Strain Effects on Heat Transport in Silicene — Poster @ **APS March Meeting** (2025)
- Mode Localization and Suppressed Heat Transport in Amorphous Mats. — **Physical Review B** (2021)
- Efficient Anharmonic Lattice Dynamics Calculations — **Journal of Applied Physics** (2020)

AWARDS & AFFILIATIONS

Representative @ Chemistry Graduate Student Association, Physical Track (2024)

Representative @ UC Academic Senate Committee on Information Technology (2023)

Quantum Multi-scale Modeling Workshop (2021)

93rd percentile — GRE Verbal Reasoning (2018)

96th percentile — ACS Organic Chemistry National Exam (2018)

Rotaract Service Above Self Award

Four consecutive Dean's List honors